



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 3

Assignment: Modelling Language – UML

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 3 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 3 module and your lecture notes before attempting each question.
- Draw diagrams neatly (by hand or using a UML/diagramming tool) with correct notation.
- Cite OMG UML 2.5.1 and MOF where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define UML and state what each part of the name means. [3 marks]

Question A2

Explain the difference between a modelling language and a development methodology, using UML and UP as examples. [2 marks]

Question A3

Name the three conceptual building blocks of UML. [2 marks]

Question A4

List and give one example each of the four kinds of UML 'things'. [2 marks]

Question A5

Name the four main UML relationship types. [2 marks]

Question A6

Differentiate between syntax and semantics in UML. [2 marks]

Question A7

Explain the four common mechanisms of UML. [2 marks]

Question A8

Define a stereotype, a tagged value and a constraint, with an example of each. [3 marks]

Question A9

Explain the four UML layers M3, M2, M1 and M0. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

A university wants a mobile student registration application. Students can log in, view courses, register courses, drop courses and view their registration status. The application communicates with a university server and a database.

For each requirement below, state the most appropriate UML diagram and justify your choice:

- (a) Show what functions students can perform. [2 marks]
- (b) Show the Student and Course classes and their relationship. [2 marks]
- (c) Show the login communication between the app, server and database. [2 marks]
- (d) Show the steps in the course-registration workflow. [2 marks]
- (e) Show the states of a registration (Draft → Submitted → Approved). [2 marks]
- (f) Show where the app, server and database are deployed. [2 marks]

Question B2 (8 marks)

Draw a simple UML class diagram for the university registration system containing Student, Course, Lecturer and Registration.

- (a) Show the four classes with two attributes each. [2 marks]
- (b) Add the relationships Student—Registration, Course—Registration and Lecturer—Course. [2 marks]
- (c) Add sensible multiplicities to two of the relationships. [2 marks]
- (d) Add one constraint or note expressing a business rule (e.g. prerequisites). [2 marks]

Section C — Case Study (10 marks)

'An online banking system allows customers to view balances, transfer money between accounts and pay bills. The system authenticates customers, communicates with a core banking server and a database, and must encrypt all transfers.'

As an object-oriented analyst:

- (a) Explain how UML would help the analyst, developer, database designer and bank manager communicate about this system. [3 marks]
- (b) Identify which UML diagrams you would use to show: the functions a customer can perform; the Account and Transaction classes; the order of a transfer; the states of a transaction. [4 marks]
- (c) Use the four-layer architecture (M0–M3) to explain where 'Customer', 'Account', 'Class' and 'MOF' belong. [3 marks]



Marking Scheme

Section	Content	Marks
A	Short-answer questions (9 questions)	20
B	Application questions (12 + 8)	20
C	Case study	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Cite any sources you consult using the format used in the Week 3 module references.